

Engineering Portfolio: High-Power Audio Amplifier

PCB Design, Routing Optimization, & Verification Project

Designer: Kouassi Ngoran Abel | **Course:** EET 414 | **Institution:** University of Maine |
Date: April 26, 2026

Project Executive Summary

This engineering portfolio integrates the complete hardware design cycle of a high-fidelity power audio amplifier implemented on a dual-layer PCB structure using Altium Designer. The system features a low-noise input op-amp stage (LF341) coupled with high-current complementary push-pull Darlington transistors (TIP122/TIP127) optimized to drive an 8-ohm loudspeaker load safely and efficiently.

Board Design & Physical Metrics

- **Compact Form Factor:** 975.00 mil × 775.00 mil (0.975 in × 0.775 in)
- **Total Surface Footprint Area:** 0.755 sq. in.
- **Layer Topology:** 2-Layer rigid copper array with a dedicated polygon ground and -12V distribution planes.

Technical Reference Matrix

Designator	Component Specification	Engineering Responsibility
U1	LF341 Operational Amplifier	Primary voltage gain & signal isolation stage.
Q1 / Q2	TIP122 (NPN) / TIP127 (PNP) Pairs	Complementary push-pull active high-current power stage.
C1 / C2	10pF Ceramic / 33μF Electrolytic	Decoupling and high-frequency noise filtering across supply rails.

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Date: 04/26/2026



University of Maine

Spring 2026

Final Project_High_power_audio_amplifier.

The final project of high power audio amplifier is given in the following screenshot.

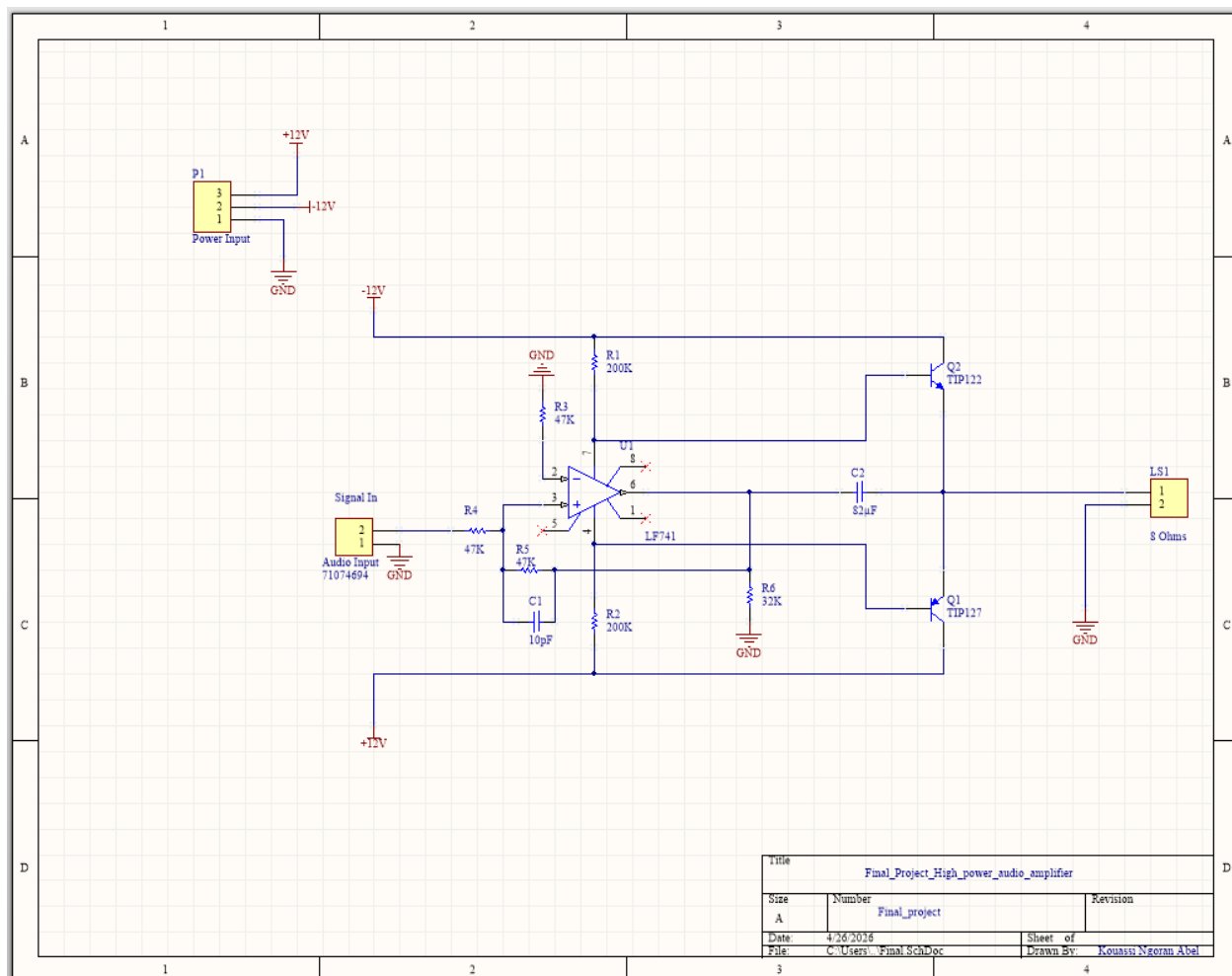


Figure 1: Schematic of High-power audio amplifier.

Design Rule Verification Report

Date : 4/26/2026
 Time : 9:18:26 PM
 Elapsed Time : 00:00:01
 Filename : [C:\Users\Public\Documents\Altium\Projects\AudioAmp_Flat\Final\Final.PcbDoc](#)

Warnings : 0
Rule Violations : 0

Summary

Warnings	Count
Total	0
Rule Violations	Count
Modified Polygon (Allow modified: No), (Allow shelved: No)	0
Net Antennae (Tolerance=0mil) (All)	0
Silk primitive without silk layer	0
Silk to Silk (Clearance=10mil) (Disabled)(All),(All)	0
Silk To Solder Mask (Clearance=10mil) (Disabled)(IsPad),(All)	0
Minimum Solder Mask Sliver (Gap=10mil) (Disabled)(All),(All)	0
Hole To Hole Clearance (Gap=10mil) (All),(All)	0
Hole Size Constraint (Min=1mil) (Max=100mil) (All)	0
Height Constraint (Min=0mil) (Max=1000mil) (Preferred=500mil) (All)	0
Width Constraint (Min=8mil) (Max=8mil) (Preferred=8mil) (All)	0
Power Plane Connect Rule(Direct Connect) (Expansion=20mil) (Conductor Width=10mil) (Air Gap=10mil) (Entries=4) (All)	0
Clearance Constraint (Gap=8mil) (All),(All)	0
Un-Routed Net Constraint ((All))	0
Short-Circuit Constraint (Allowed=No) (All),(All)	0
Total	0

Figure 2: Design rules Summary of high power audio amplifiers.

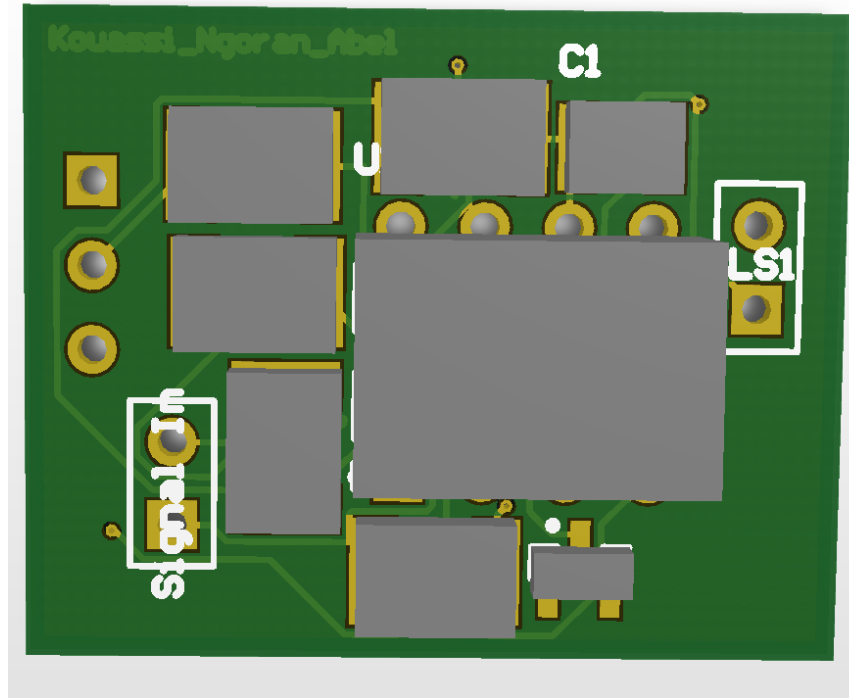


Figure 3: 3-D Top view of high-power audio amplifier.

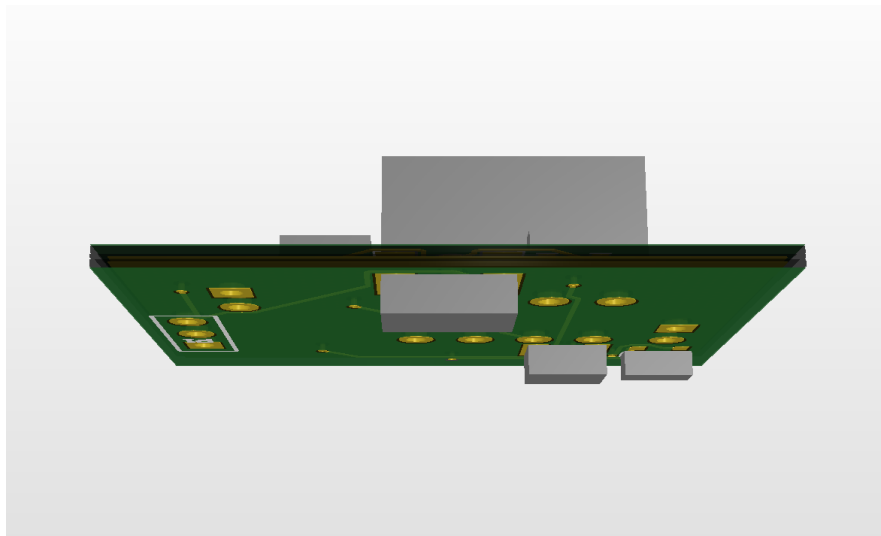


Figure 4: 3-D Bottom view of high-power audio amplifier.

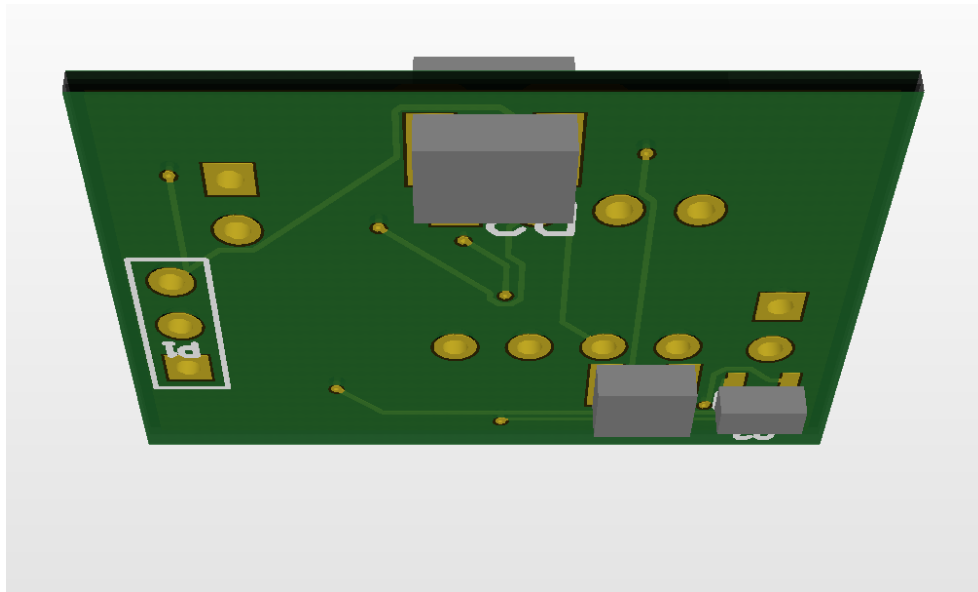


Figure 5 : 3-D Bottom view of high-power audio amplifier. (full view)

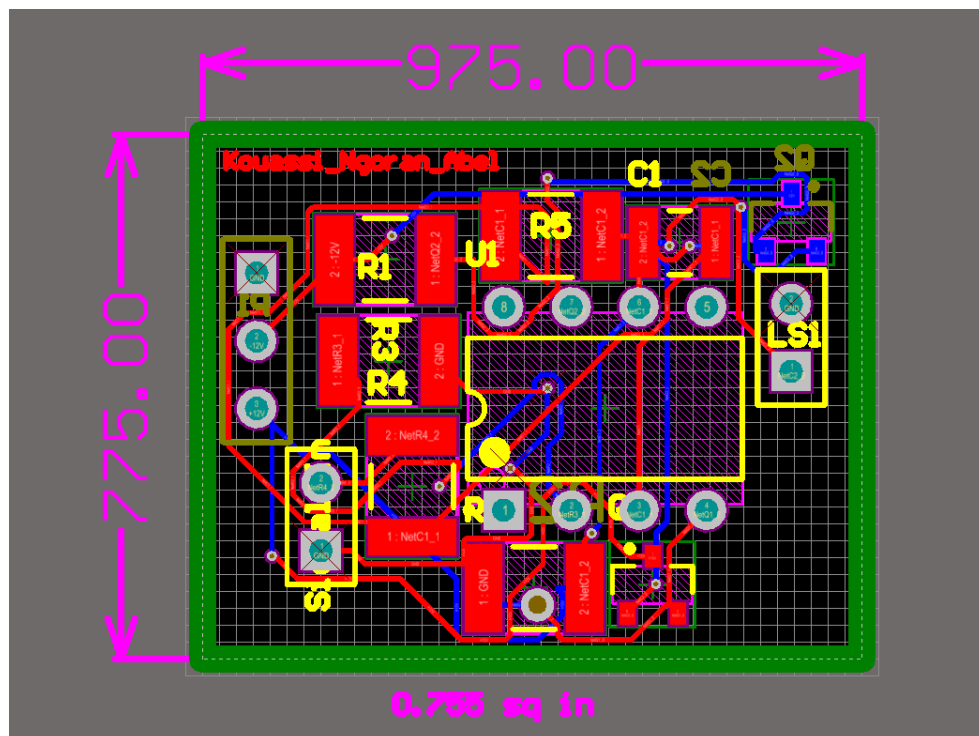


Figure 6: Top layer

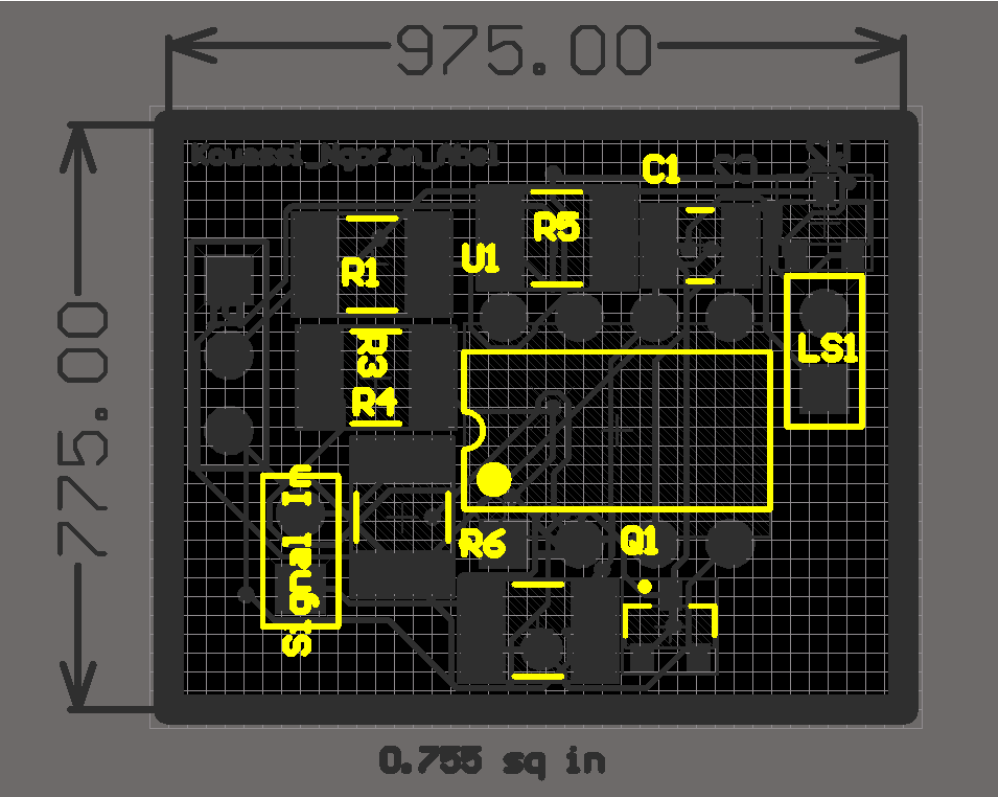


Figure 7: Shift S top Overlay

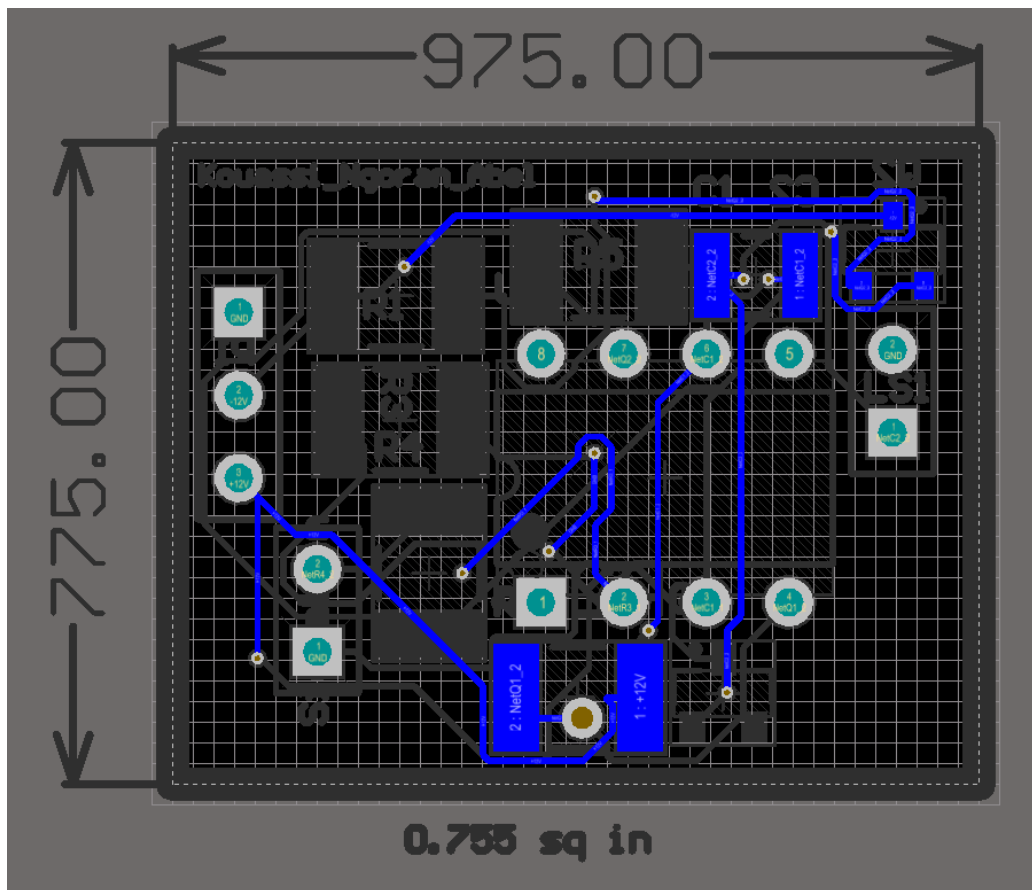


Figure 9: Shift S Bottom

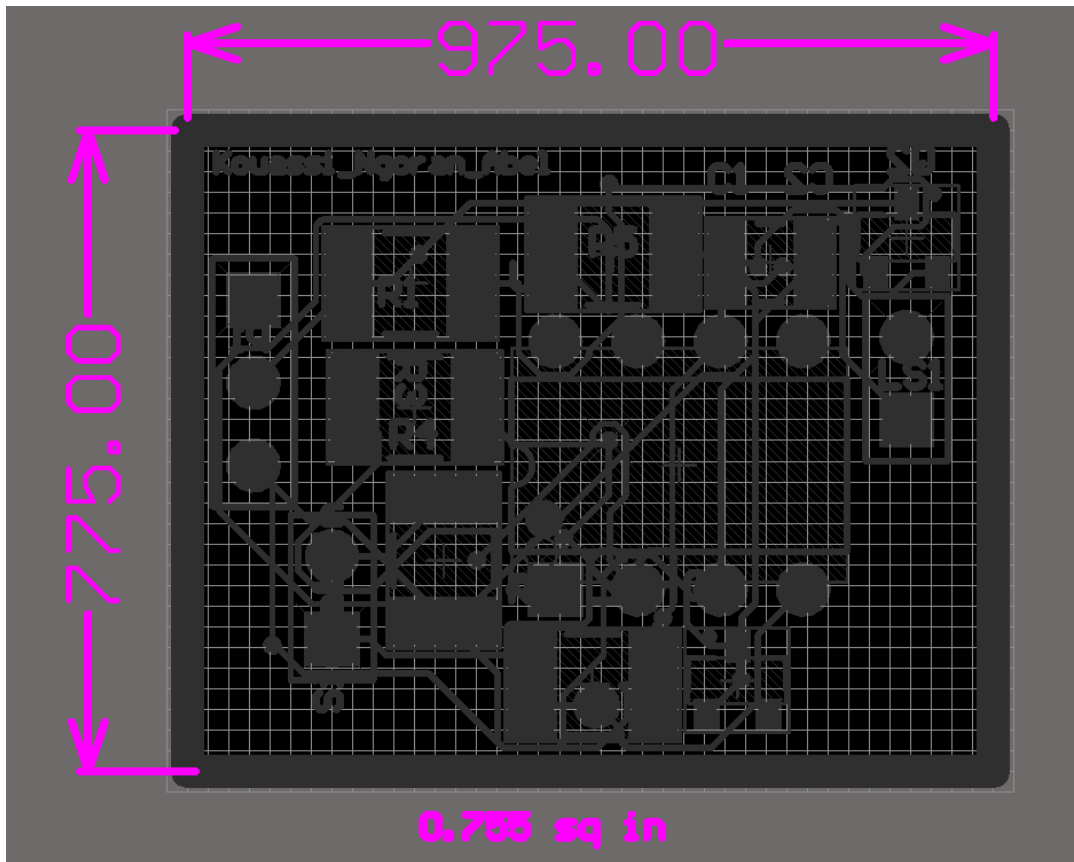


Figure 10: Shift S Mechanical 1

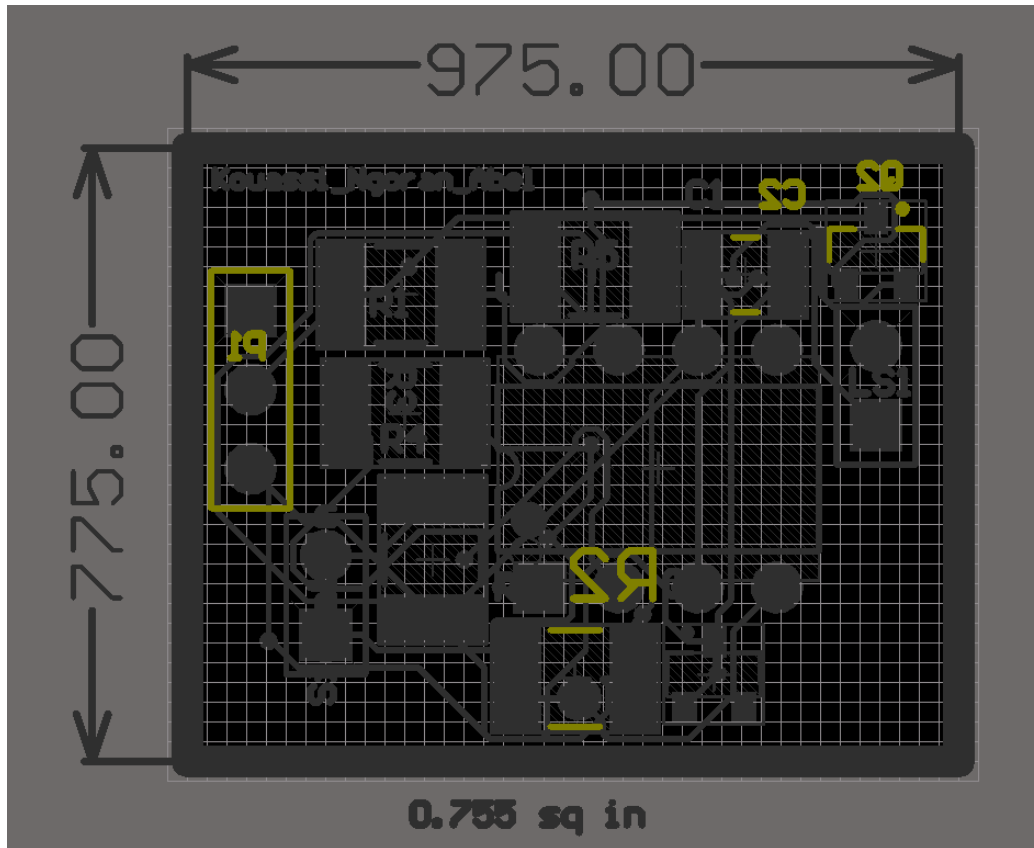


Figure 11: Shift S Bottom Overlay

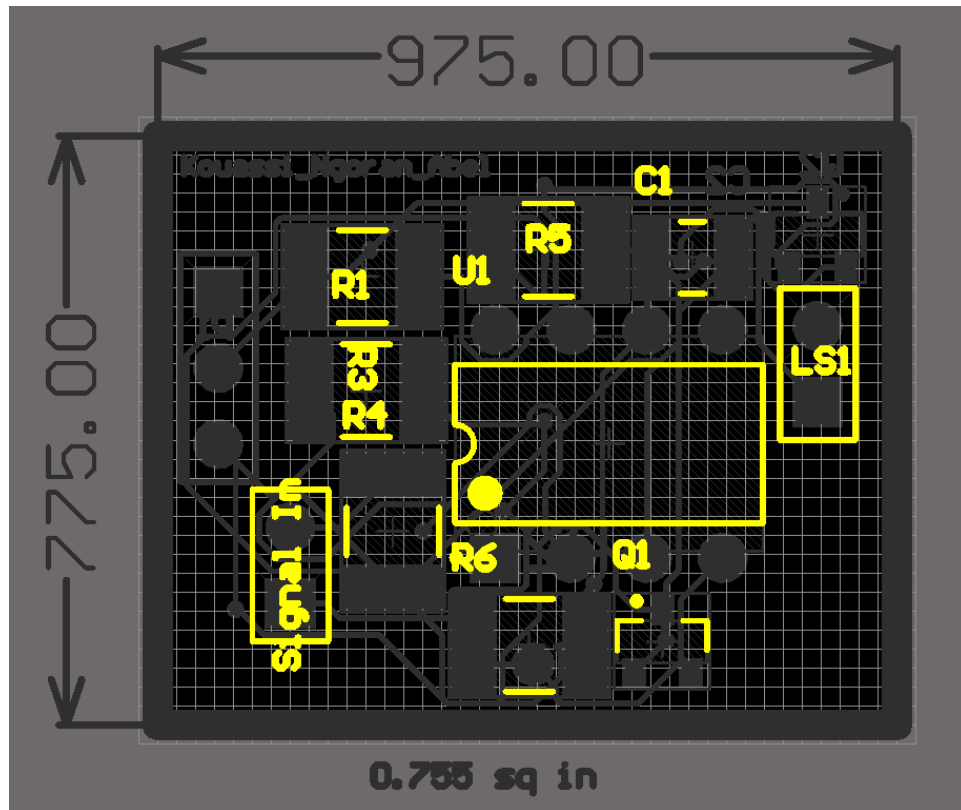


Figure 12: Shift S Top Overlay

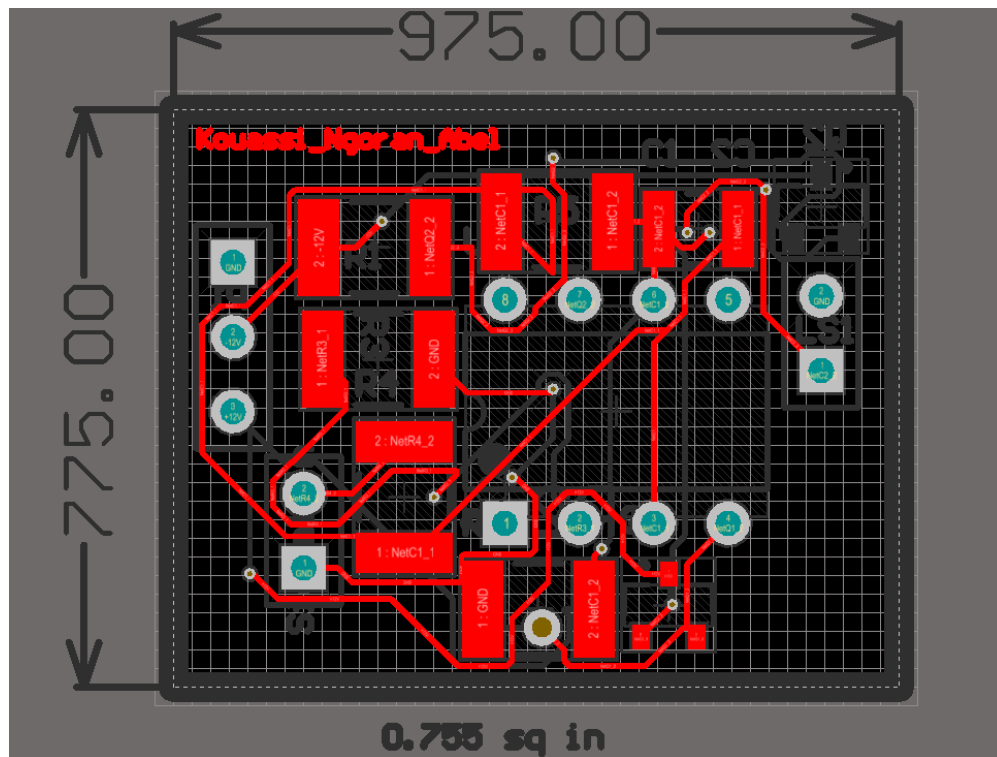


Figure 13: Shift S Top layer

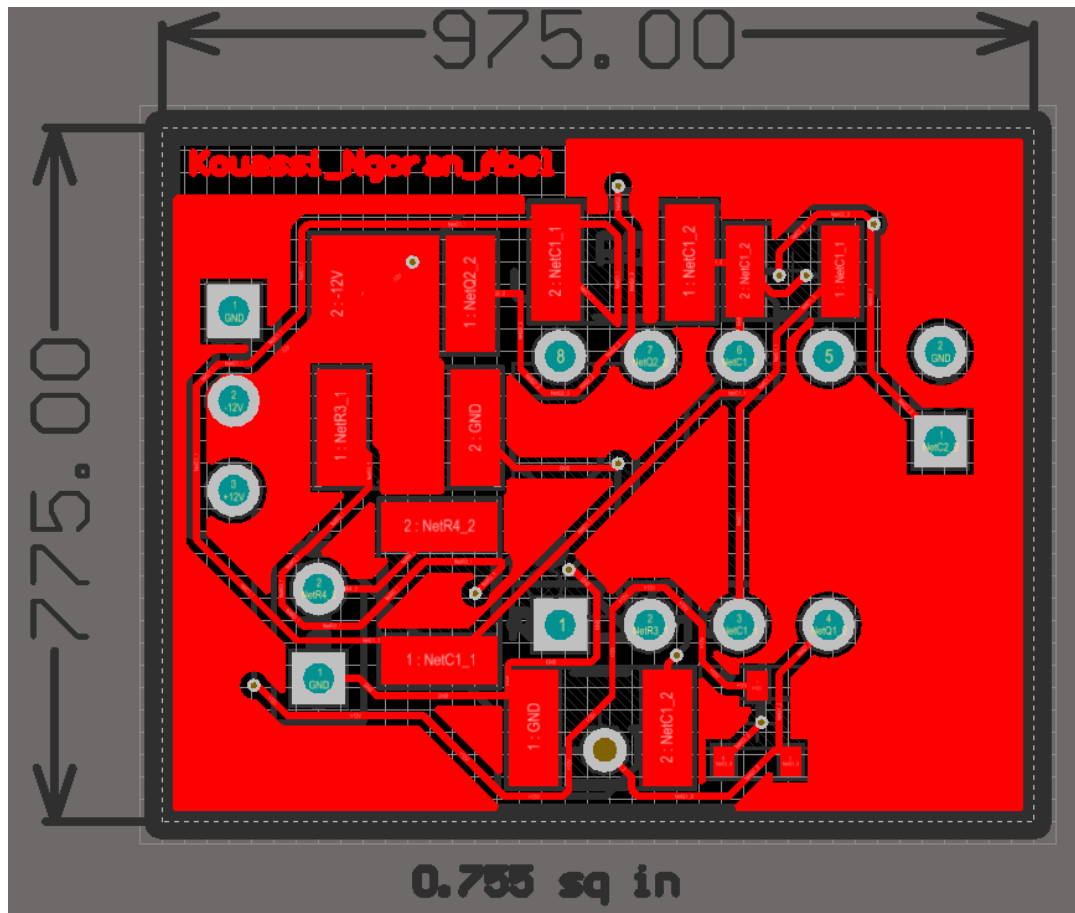


Figure 14: Shift S Top layer

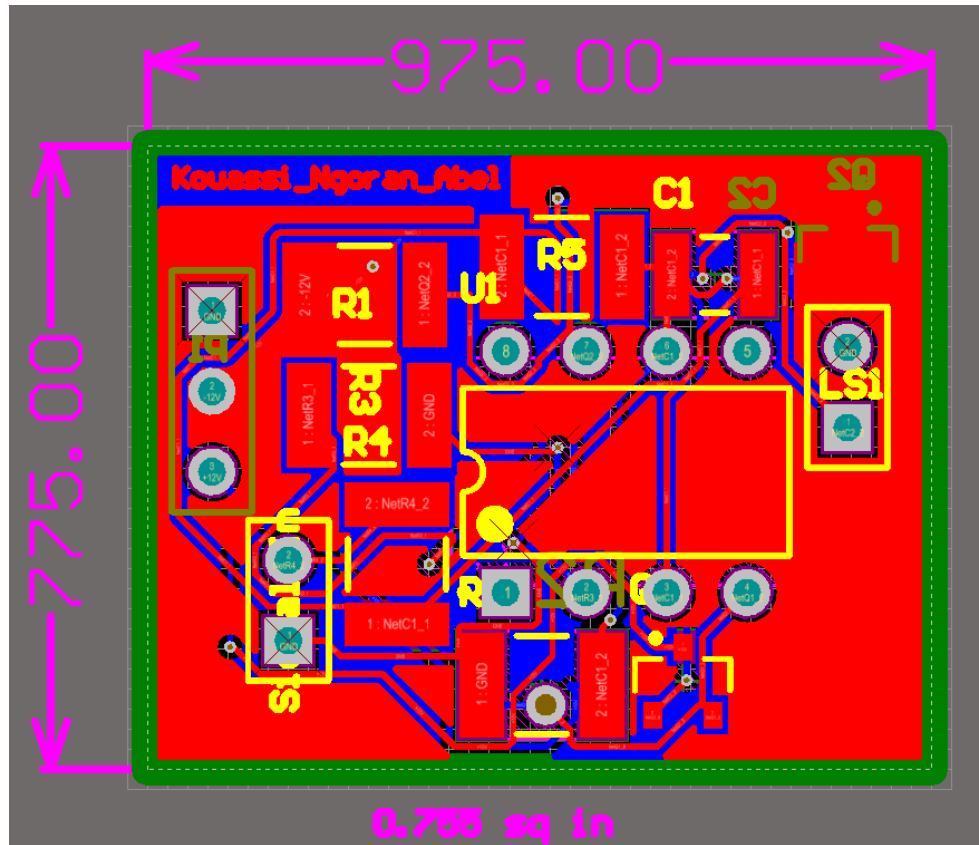


Figure 15: Top layer with copper (-12V)

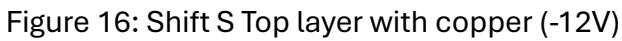


Figure 16: Shift S Top layer with copper (-12V)

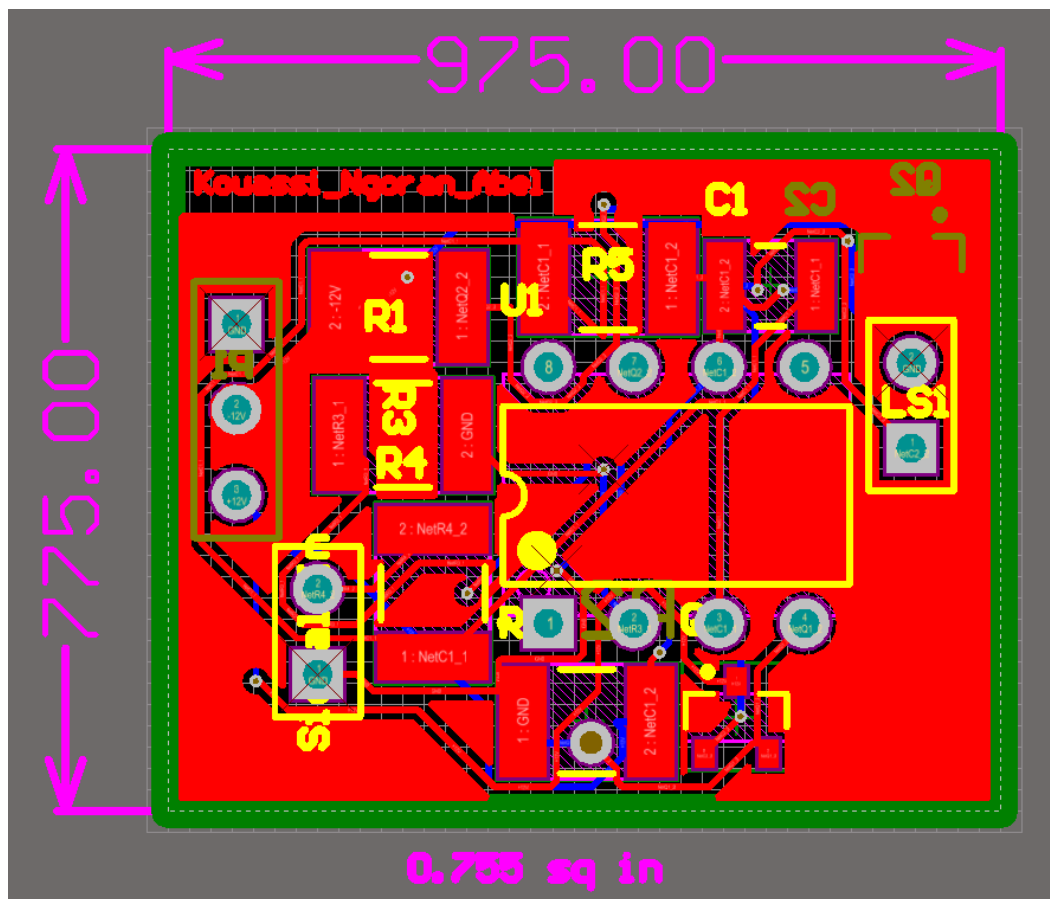


Figure 17: Top Layer copper (-12V)

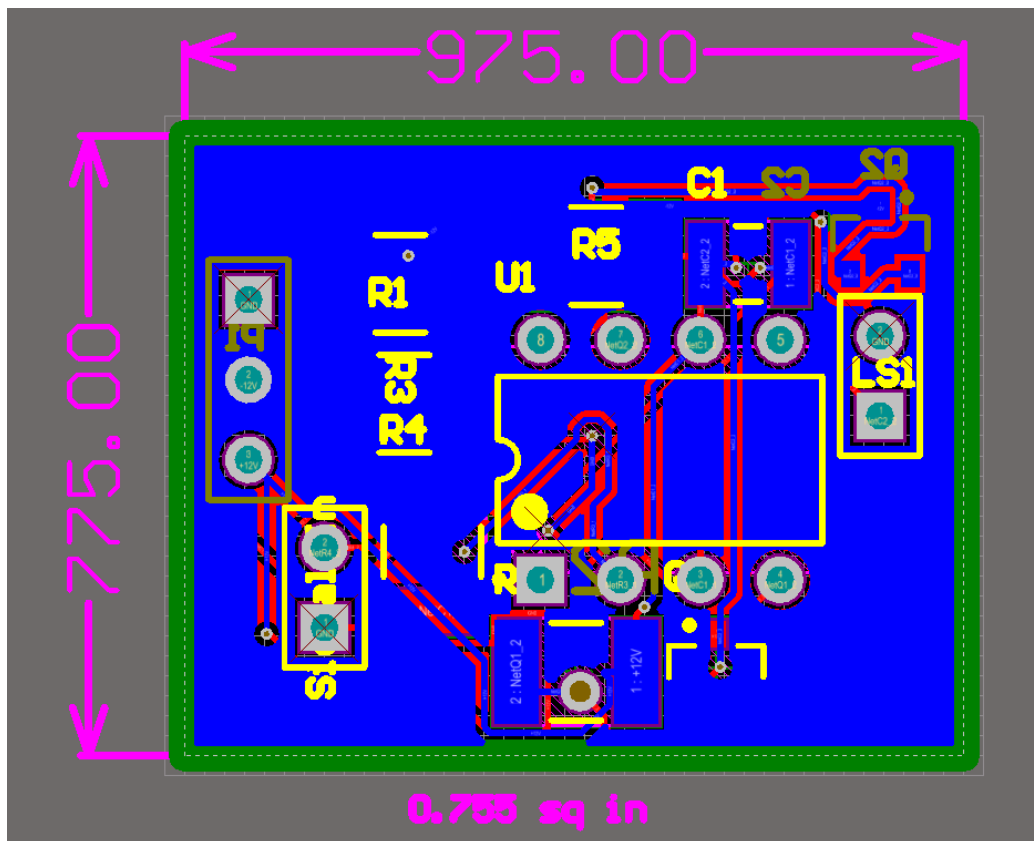


Figure 18 : Bottom Layer with copper (-12V)

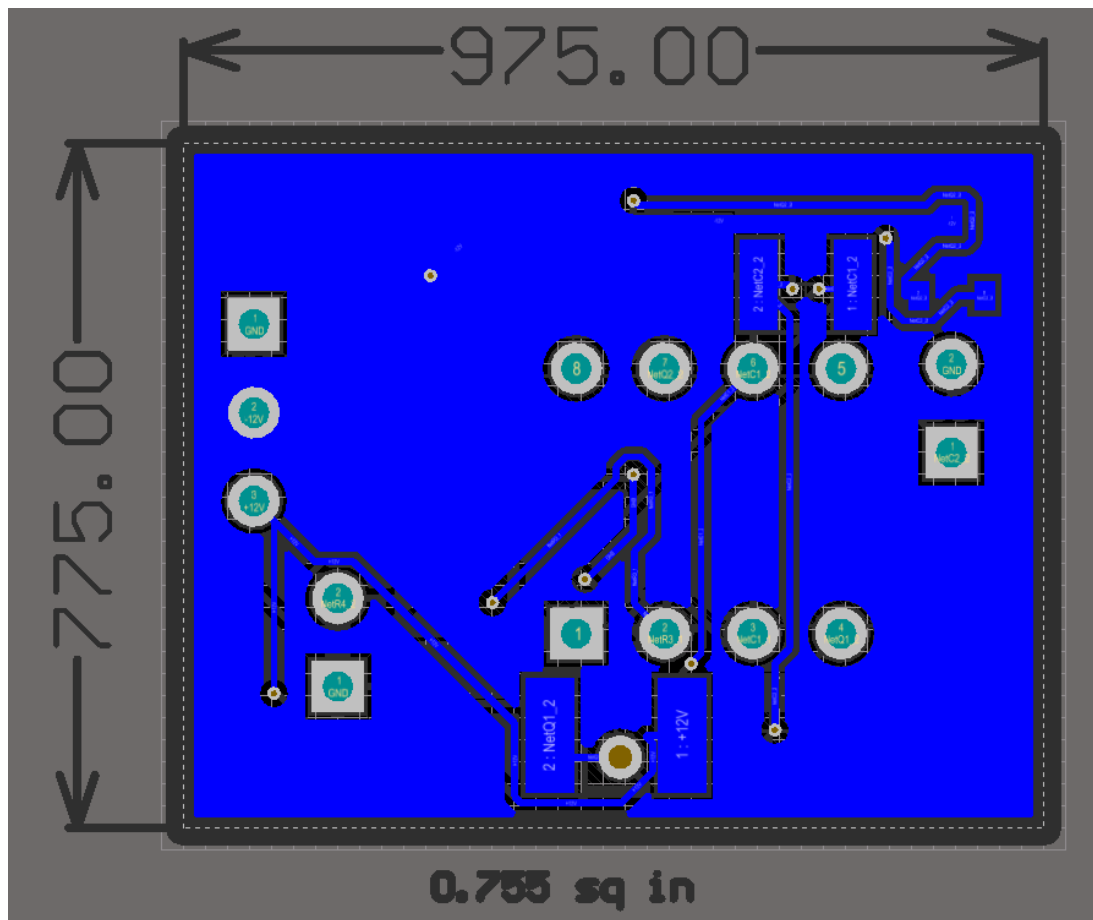


Figure 19 : Shift S Bottom Layer with Copper (-12V)

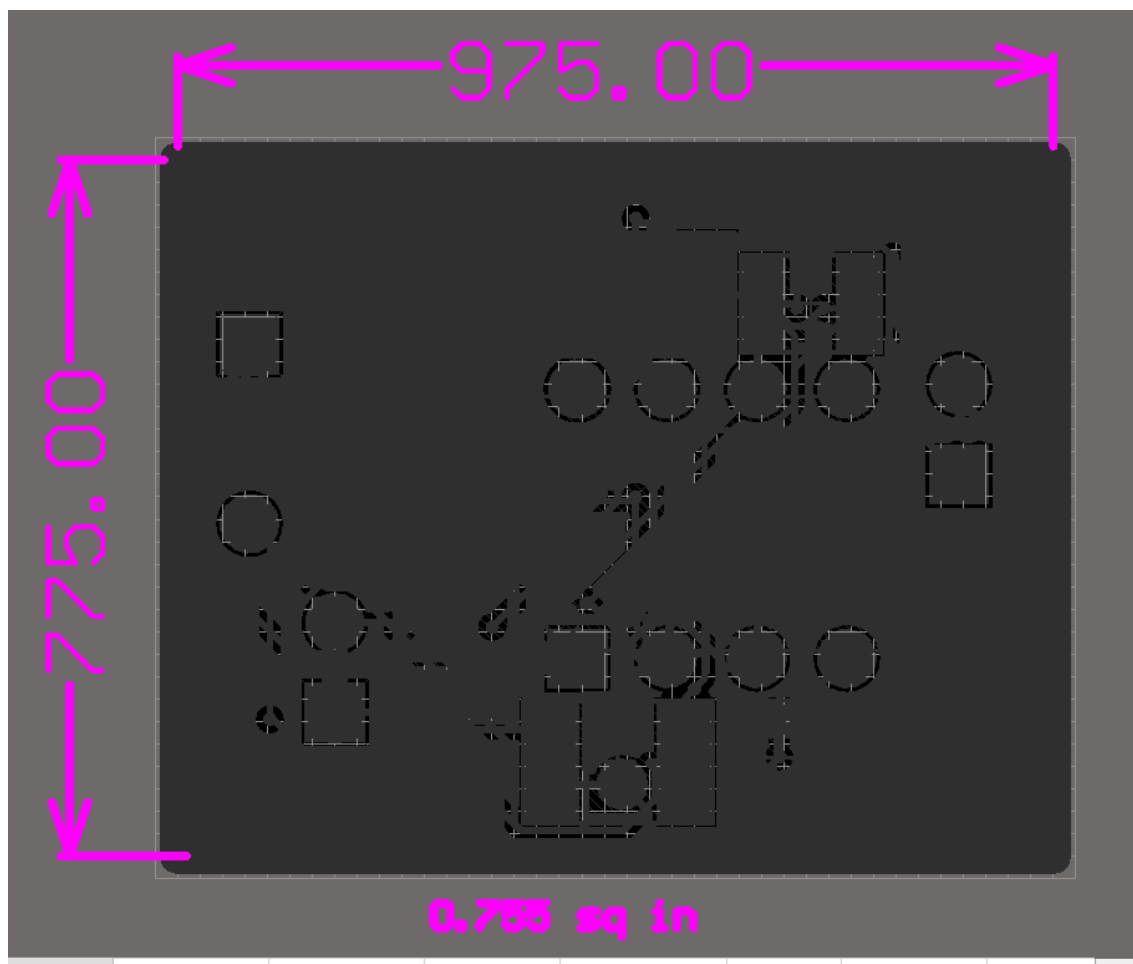


Figure 20: Mechanical 1

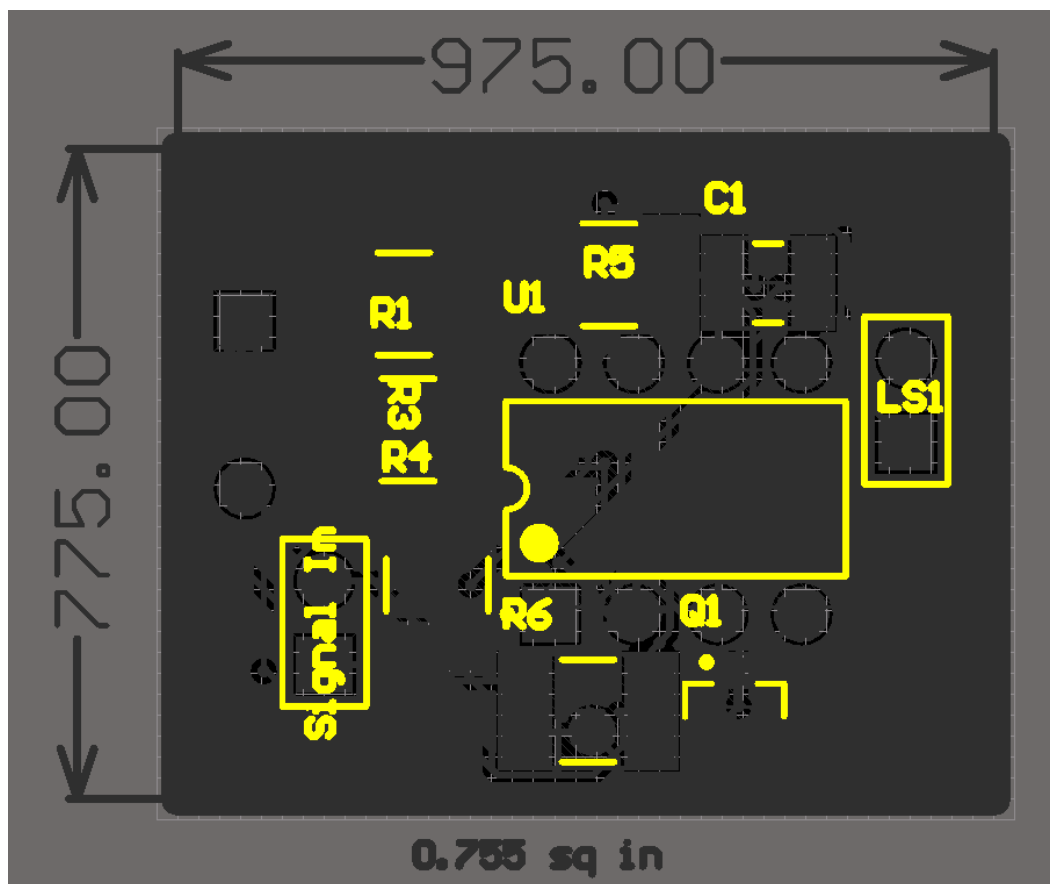


Figure 21: Top Overlay

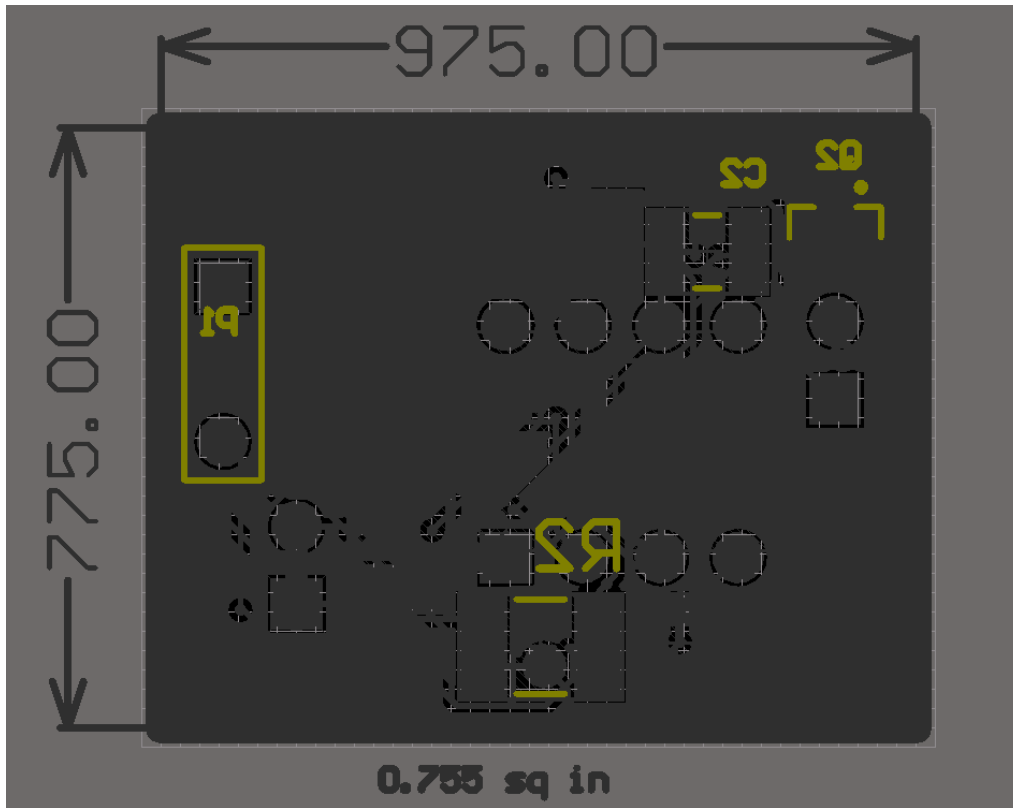


Figure 22: Bottom Overlay

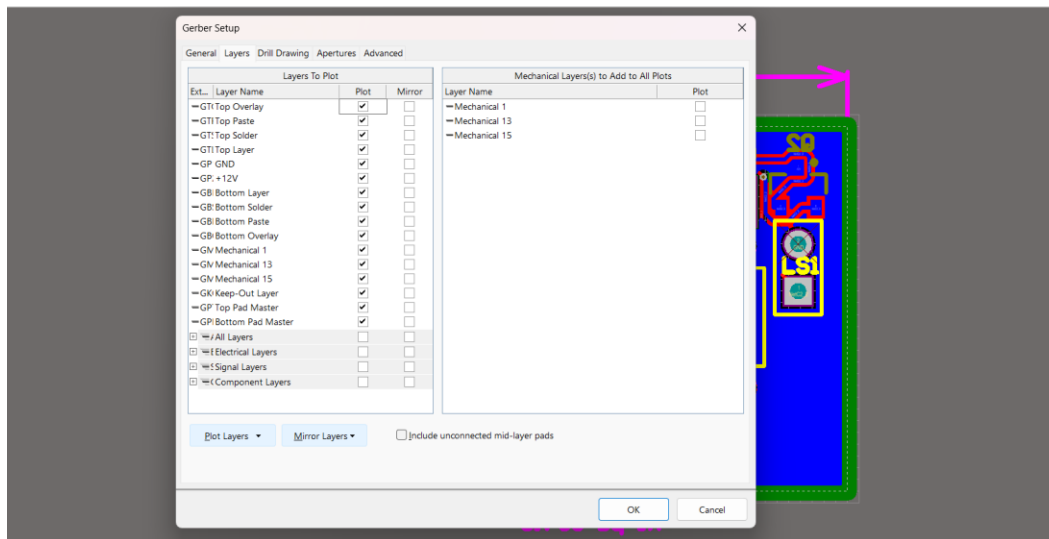


Figure 23: Gerber Setup

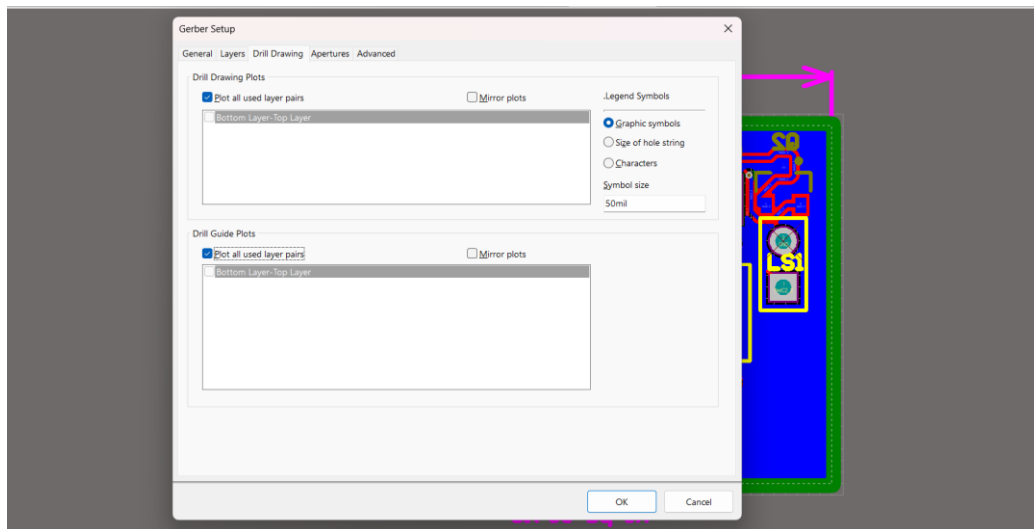


Figure 24 : Gerber Setup



Figure 25 .

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Design Rule Check - Final	4/26/2026 10:09 PM	Microsoft Edge HTML Document	12 KB
Final	4/26/2026 11:06 PM	CAMtastic Aperture Data	3 KB
Final	4/26/2026 11:11 PM	Altium NC Drill Report File	1 KB
Final.EXTREP	4/26/2026 11:06 PM	EXTREP File	2 KB
Final	4/26/2026 11:06 PM	CAMtastic Bottom Layer Gerber Data	13 KB
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Figure 26: Camtastic files.